

Distinct Routes

Time limit: 1.00 s

Memory limit: 512 MB

A game consists of n rooms and m teleporters. At the beginning of each day, you start in room 1 and you have to reach room n .

You can use each teleporter at most once during the game. How many days can you play if you choose your routes optimally?

Input

The first input line has two integers n and m : the number of rooms and teleporters. The rooms are numbered $1, 2, \dots, n$.

After this, there are m lines describing the teleporters. Each line has two integers a and b : there is a teleporter from room a to room b .

There are no two teleporters whose starting and ending room are the same.

Output

First print an integer k : the maximum number of days you can play the game. Then, print k route descriptions according to the example. You can print any valid solution.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq a, b \leq n$

Example

Input	Output
6 7 1 2 1 3 2 6 3 4 3 5 4 6 5 6	2 3 1 2 6 4 1 3 4 6